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Cambridge Ordinary Level

MARK SCHEME for the October/November 2014 series

4037 ADDITIONAL MATHEMATICS

4037/12 Paper 1, maximum raw mark 80

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1	$\frac{dy}{dx} = 2x - \frac{16}{x^2}$ When $\frac{dy}{dx} = 0$, $x = 2, y = 12$	**M1** **A1** **DM1** **A1**	for attempt to differentiate all correct for equating $\frac{dy}{dx}$ to zero and an attempt to solve for x. **A1** for both, but no extra solutions
2 **(a)**		**B1** **B1** **B1**	for correct shape for max value of 2, starting at (0, 2) and finishing at (180° , 2) for min value of –4
(b) (i)	4	**B1**	must be positive
(ii)	60° or $\frac{\pi}{3}$ or 1.05 rad	**B1**	
3 **(i)**	$y = 4(x+3)^{\frac{1}{2}}(+c)$ $10 = 4\left(9^{\frac{1}{2}}\right) + c$ $c = -2$ $y = 4(x+3)^{\frac{1}{2}} - 2$	**M1, A1** **M1** **A1**	**M1** for $(x+3)^{\frac{1}{2}}$, **A1** for $4(x+3)^{\frac{1}{2}}$ for a correct attempt to find c, but must be from an attempt to integrate Allow **A1** for $c = -2$
(ii)	$6 = 4(x+3)^{\frac{1}{2}} - 2$ $x = 1$	**A1 ft**	**ft** for substitution into *their* equation to obtain x; must have the first **M1**

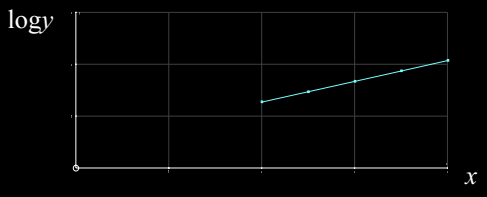
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4	**(i)**	$5y^2 - 7y + 2 = 0$	**B1, B1**	**B1** for 5, **B1** for –7
	(ii)	$(5y - 2)(y - 1) = 0$	**M1**	for solution of quadratic equation from **(i)**
		$y = \frac{2}{5}, x = \frac{\ln 0.4}{\ln 5}$	**M1**	for use of logarithms to solve equation of the type $5^x = k$
		$x = -0.569$	**A1**	must be evaluated to 3sf or better
		$y = 1, x = 0$	**B1**	
5	**(i)**	$\frac{dy}{dx} = 3x^2 - \frac{1}{x}$	**M1**	for attempt to differentiate
		When $x = 1$, $y = 1$ and $\frac{dy}{dx} = 2$	**B1**	for $y = 1$
		Tangent: $y - 1 = 2(x - 1)$	**DM1**	for attempt to find equation of tangent
		$(y = 2x - 1)$	**A1**	allow equation unsimplified
	(ii)	Mid-point (5, 9)	**B1**	for midpoint from given coordinates
		$9 = 2(5) - 1$	**B1**	for checking the mid-point lies on tangent
		Alternative Method: Tangent equation $y = 2x - 1$ Equation of line joining (–2, 16) and (12, 2) $y = -x + 14$ Solve simultaneously $x = 5$, $y = 9$	**B1**	for a complete method to find the coordinates of the point of intersection
		Mid-point (5, 9)	**B1**	for midpoint from given coordinates
6	**(i)**	$(2 + px)^6 = 64 + 192px + 240p^2x^2 \ldots$	**B1**	for $240p^2$ or $240p^2x^2$ or ${}^6C_2 \times 2^4 \times (px)^2$ or ${}^6C_2 \times 2^4 \times p^2$ or ${}^6C_2 \times 2^4 \times p^2x^2$
		$240p^2 = 60$	**M1**	for equating *their* term in x^2 to 60 and attempt to solve
		$p = \frac{1}{2}$	**A1**	
	(ii)	$(3 - x)(64 + 192px + 240p^2x^2 \ldots)$	**B1 ft**	**ft** for $192p$, 96 or $192 \times$ *their* p
		Coefficient of x^2 is $180 - 192p$	**M1**	for $180 - 192p$
		= 84	**A1**	

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7	**(i)**	$\mathbf{A}^{-1} = \frac{1}{5ab}\begin{pmatrix} b & -2b \\ a & 3a \end{pmatrix}$	**B1, B1**	**B1** for $\frac{1}{5ab}$, **B1** for $\begin{pmatrix} b & -2b \\ a & 3a \end{pmatrix}$
	(ii)	$\mathbf{X} = \mathbf{BA}^{-1}$	**M1**	for post-multiplication by inverse matrix
		$= \begin{pmatrix} -a & b \\ 2a & 2b \end{pmatrix}\begin{pmatrix} \frac{1}{5a} & -\frac{2}{5a} \\ \frac{1}{5b} & \frac{3}{5b} \end{pmatrix}$	**DM1**	for correct attempt at matrix multiplication, needs at least one term correct for their BA^{-1} (allow unsimplified)
		$= \begin{pmatrix} 0 & 1 \\ \frac{4}{5} & \frac{2}{5} \end{pmatrix}$	**A1** **A1**	for each correct pair of elements, must be simplified
8	**(i)**	$\overrightarrow{AB} = \begin{pmatrix} 12 \\ 16 \end{pmatrix}$, at P, $x = -2 + \frac{1}{4}(12)$ so at P, $x = 1$	**B1**	for convincing argument for $x = 1$
		$y = 3 + \frac{1}{4}(16)$, $y = 7$	**B1**	for $y = 7$
	(ii)	Gradient of $AB = \frac{16}{12}$, so perp gradient = $-\frac{3}{4}$	**M1**	for finding gradient of perpendicular
		Perp line: $y - 7 = -\frac{3}{4}(x - 1)$ $(3x + 4y = 31)$	**M1** **A1**	for equation of perpendicular through their P Allow unsimplified
	(iii)	$Q\left(0, \frac{31}{4}\right)$	**B1 ft** **M1**	**ft** on their perpendicular line, may be implied for any valid method of finding the area of the correct triangle, allow use of *their Q*; must be in the form $(0, q)$.
		Area $AQB = 12.5$	**A1**	

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9 **(i)**	$\log y = \log a + x \log b$	**B1**	for the statement, may be seen or implied in later work,
	x: 2, 2.5, 3, 3.5, 4 lg y: 1.27, 1.47, 1.67, 1.87, 2.07		
	ln$y$: 2.93, 3.39, 3.84, 4.31, 4.76		
		**M1** **A2,1,0**	for attempt to draw graph of x against log y –1 each error in points plotted
(ii)	Gradient = log b $\lg b = 0.4$ or $\ln b = 0.92$	**DM1**	for attempt to find gradient and equate it to log b, dependent on **M1** in **(i)**
	$b = 2.5$ (allow 2.4 to 2.6)	**A1**	
	Intercept = log a $\lg a = 0.47$ or $\ln a = 1.10$	**DM1**	for attempt to equate y-intercept to log a or use *their* equation with *their* gradient and a point on the line, dependent on **M1** in **(i)**
	$a = 3$ (allow 2.8 to 3.2)	**A1**	
	Alternative method: Simultaneous equations may be used provided points that are on the plotted straight line are used.	**DM1** **DM1**	for a pair of equations using points on the line, dependent on **M1** in **(i)** for solution of these equations, dependent on **M1** in **(i)**
	$a = 3$ (allow 2.8 to 3.2) $b = 2.5$ (allow 2.4 to 2.6)	**A1** **A1**	**A1** for each

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10 (a) (i)	360	**B1**	
(ii)	60	**B1**	
(iii)	36	**B1**	
(b) (i)	${}^{8}C_5 \times {}^{12}C_5$	**B1, B1**	**B1** for each, allow unevaluated with no extra terms
	$56 \times 792 = 44352$	**B1**	Final answer must be evaluated and from multiplication
(ii)	4 places are accounted for Gender no longer 'important'	**M1**	for realising that 4 places are accounted or that gender is no longer important
	Need ${}^{16}C_6 = 8008$	**A1**	for 8008
	Alternative Method $\left({}^{6}C_6 \times {}^{10}C_0\right) + \left({}^{6}C_5 \times {}^{10}C_1\right) \ldots \left({}^{6}C_0 \times {}^{10}C_6\right)$ $1 + 60 + 675 + 2400 + 3150 + 1512 + 210 = 8008$	**M1** **A1**	for at least 5 of the 7 cases, allow unsimplified
11 (a)	$2\cos 3x - \dfrac{\cos 3x}{\sin 3x} = 0$ $\cos 3x\left(2 - \dfrac{1}{\sin 3x}\right) = 0$	**M1**	for use of $\cot 3x = \dfrac{\cos 3x}{\sin 3x}$, may be implied
	Leading to $\cos 3x = 0,\ 3x = 90^\circ,\ 270^\circ$	**DM1**	for attempt to solve $\cos 3x = 0$ correctly from correct factorisation to obtain x
	$x = 30^\circ, 90^\circ$	**A1**	**A1** for both, no excess solutions in the range
	and $\sin 3x = \dfrac{1}{2},\ 3x = 30^\circ,\ 150^\circ$	**DM1**	for attempt to solve $\sin 3x = \dfrac{1}{2}$ correctly to obtain x
	$x = 10^\circ, 50^\circ$	**A1**	**A1** for both, condone excess solutions
(b)	$\cos\left(y + \dfrac{\pi}{2}\right) = -\dfrac{1}{2}$ $y + \dfrac{\pi}{2} = \dfrac{2\pi}{3}, \dfrac{4\pi}{3}$	**M1**	for dealing with sec $\left(y + \dfrac{\pi}{2}\right)$ correctly
		DM1	for correct order of operations, must not mix degrees and radians
	so $y = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$ (0.524, 2.62)	**A1, A1**	

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12	**(i)**	$\overrightarrow{AQ} = \lambda\mathbf{b} - \mathbf{a}$	**B1**	
	(ii)	$\overrightarrow{BP} = \mu\mathbf{a} - \mathbf{b}$	**B1**	
	(iii)	$\overrightarrow{OR} = \mathbf{a} + \frac{1}{3}(\lambda\mathbf{b} - \mathbf{a})$ or $\lambda\mathbf{b} - \frac{2}{3}(\lambda\mathbf{b} - \mathbf{a})$	**M1**	for $\mathbf{a} + \frac{1}{3}$ *their* **(i)**
		$= \frac{2}{3}\mathbf{a} + \frac{1}{3}\lambda\mathbf{b}$	**A1**	Allow unsimplified
	(iv)	$\overrightarrow{OR} = \mathbf{b} + \frac{7}{8}(\mu\mathbf{a} - \mathbf{b})$ or $\mu\mathbf{a} - \frac{1}{8}(\mu\mathbf{a} - \mathbf{b})$	**M1**	for $\mathbf{b} + \frac{7}{8}$ *their* **(ii)**
		$= \frac{1}{8}\mathbf{b} + \frac{7}{8}\mu\mathbf{a}$	**A1**	Allow unsimplified
	(v)	$\frac{2}{3}\mathbf{a} + \frac{1}{3}\lambda\mathbf{b} = \frac{1}{8}\mathbf{b} + \frac{7}{8}\mu\mathbf{a}$	**M1**	for equating **(iii)** and **(iv)** and then equating like vectors
		$\frac{2}{3} = \frac{7}{8}\mu, \mu = \frac{16}{21}$ Allow 0.762	**A1**	
		$\frac{1}{3}\lambda = \frac{1}{8}, \lambda = \frac{3}{8}$ Allow 0.375	**A1**	

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